

LLM-powered Text Entry in Virtual Reality

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ABSTRACT

Large language models (LLMs) have demonstrated exceptional performance across various language-related tasks, offering significant potential for enhancing text entry in Virtual Reality (VR). We introduce an LLM-powered text entry system for VR, which integrates multiple input modalities and utilizes a fine-tuned LLM as a keyboard decoder. The LLM-based decoder achieved 93.1% top-1 decoding accuracy on a word gesture typing dataset and 95.4% on tap typing, highlighting its potential for VR text entry applications. Our demonstration shows how LLMs can support tap typing and word-gesture typing through raycasting and joystick-based inputs, potentially accommodating various user preferences and enhancing the adaptability of VR text input methods.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Text entry; Human-centered computing—Human computer interaction (HCI)—Virtual reality.

1 INTRODUCTION

Large language models (LLMs) have shown significant advantages across a broad range of language-related tasks [1], including text generation and summarization, which led to increased exploration in their diverse applications. The potential of LLMs spans traditional text-based domains to new areas, such as keyboard decoding [7], which involves converting input signals—such as touch points or gesture trajectories—into words. However, text entry in Virtual Reality (VR) is more complex, involving a range of input signals that go beyond conventional physical keyboards or touch-based input on smartphone soft keyboards. For instance, users can enter text using head movements [8], joystick inputs [9], or raycasting from a handheld controller [3]. To input a word, users can choose between tap typing, where they select individual letters and confirm them in sequence, and word gesture typing [5], where they draw a continuous trajectory across each letter to type the word. Given the diversity of input signals in VR, whether LLM-based decoder can effectively support keyboard decoding in this environment remains an open question. Previous work [2] has used LLMs to facilitate VR text entry through features like auto-completion, content prediction, and keyword-to-sentence generation. However, using LLMs as a keyboard decoder in VR is underexplored.

In this demo, we present an LLM-powered text entry system for VR, which enables four text input modalities: raycasting-based tap typing, raycasting-based word gesture typing, joystick-based tap typing, and joystick-based word gesture typing. Specifically, we employed a FLAN-T5 model [4], finetuned for keyboard decoding tasks, as our decoder. Input signals, such as tapped touch points or user-drawn gesture trajectories, were first transformed into letter sequences, which were then used as input for the LLM-based decoder. We developed a virtual keyboard prototype incorporating these four

input modalities to demonstrate the effectiveness of the LLM-based decoder. Our demo illustrates that the LLM-based decoder can successfully support different input signals, accommodating different user preferences and enhancing the adaptability of VR text entry.

2 TEXT INPUT MODALITIES

Our LLM-powered text entry system supports four text input modalities, each designed to accommodate diverse interaction preferences and enhance the adaptability of text input in VR.

Raycasting-based tap typing. A widely used letter-level text input method in VR, where users aim a virtual ray from a handheld controller at a target key to select it and press the controller's index trigger to confirm the selection (Fig. 1).



Figure 1: Raycasting-based tap typing.

To delete a word or select a suggestion from the suggestion bar, users direct the virtual ray from the handheld controller toward the target button and press the index trigger to confirm the action.

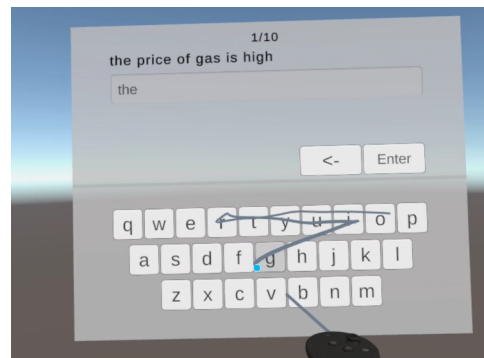


Figure 2: Raycasting-based word gesture typing.

Raycasting-based word gesture typing. Similar to raycasting-based tap typing, as shown in Fig. 2, this method allows users to aim a virtual ray from a handheld controller at a target key. To begin drawing a word gesture on the keyboard, users press and hold the controller's index trigger. While holding the trigger, users move the

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controller to guide the ray across the keyboard, creating a trajectory that represents the word gesture. The gesture is completed when users release the index trigger, signaling the end of the input. Users can also perform the same actions as in raycasting-based tap typing to delete a word or select a suggestion.

Joystick-based tap typing. A letter-level text input method where users control a on-keyboard cursor using the joystick on one handheld controller (i.e. the input controller) via a rate control mechanism. As shown in Fig. 3, to input a letter, users navigate the cursor to the desired key using the joystick and press the index trigger on the controller to confirm their selection. This process is repeated for each letter in the word, entered sequentially to input the target word.



Figure 3: Joystick-based tap typing.

Users can delete a word or select from the suggestion bar using the joystick on the other controller, called the edit controller. By swiping upward, users can select the currently focused suggestion and insert a space. A downward swipe triggers the “Backspace” function, deleting the last input signal. Swiping left or right shifts the focus between suggestions.

Joystick-based word gesture typing. Similar to joystick-based tap typing, this method involves controlling an on-keyboard cursor with the joystick on the input controller to draw word gestures on the keyboard (Fig. 4). Unlike raycasting-based word gesture typing, each word gesture begins from the same default starting cursor position, which corresponds to the neutral position of the joystick before the user starts manipulating it.



Figure 4: Joystick-based word gesture typing.

To enter a word, the user moves the cursor from its default starting position, guiding it across the letters of the intended word in sequence. Once the cursor reaches the final letter, the user releases the joystick, allowing it to return swiftly to the neutral position, ready for the next input. Users can also perform the same actions as in joystick-based tap typing to delete a word or select a suggestion.

3 PERFORMANCE OF LLM-BASED DECODER

The LLM-based decoder demonstrated strong performance on both gesture and tap typing datasets. On a real-world gesture typing dataset, it achieved a top-1 decoding accuracy of 93.1%, outperforming the *SHARK*² decoder [5], which achieved 73.2%. For top-3 accuracy, the LLM-based decoder improved the performance of *SHARK*² by 12.4%. Furthermore, the LLM-based decoder performed well on tap typing, with a top-1 accuracy of 95.4% and a top-3 accuracy of 98.6%. Building on these results, we extend the LLM-based decoder to support VR text entry.

4 INTERACTIVE DEMONSTRATION

Our demo application was developed using the Unity game engine and deployed on a Meta Quest Pro head-mounted display, with the LLM-based decoder running on an NVIDIA T4 GPU in the Google Colab cloud service. The application accesses the cloud through Unity’s web request API. The demo allows users to experience four different text entry modalities powered by a LLM-based keyboard decoder. A demo video is available online¹.

During the demo, the users can complete a transcription task with each of the four text input modalities. In the transcription task, they are asked to type a given sentence randomly selected from the MacKenzie and Soukoreff phrase set [6], with performance measured based on typing speed and word error rate. After interacting with each modality, users can view a summary of their typing performance, which helps them envision how LLMs support various input methods and enhance adaptability in VR text entry.

REFERENCES

- [1] Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, et al. A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*, 15(3):1–45, 2024. 1
- [2] L. Chen, Y. Cai, R. Wang, S. Ding, Y. Tang, P. Hansen, and L. Sun. Supporting text entry in virtual reality with large language models. In *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 524–534, 2024. doi: 10.1109/VR58804.2024.00073 1
- [3] S. Chen, J. Wang, S. Guerra, N. Mittal, and S. Prakkamakul. Exploring word-gesture text entry techniques in virtual reality. In *Extended Abstracts of the 2019 CHI conference on human factors in computing systems*, pp. 1–6, 2019. 1
- [4] H. W. Chung, L. Hou, S. Longpre, B. Zoph, Y. Tay, W. Fedus, S. Ruder, A. Chowdhery, J. Devlin, S. Levine, Y. Wang, S. S. Gu, J. Wei, T. Wolf, M. Bosma, D. Metzler, E. H. Chi, Q. V. Le, Z. Ghahramani, and O. Firat. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*, 2022. 1
- [5] P.-O. Kristensson and S. Zhai. Shark2: A large vocabulary shorthand writing system for pen-based computers. In *Proceedings of the 17th Annual ACM Symposium on User Interface Software and Technology*, UIST ’04, pp. 43–52. ACM, New York, NY, USA, 2004. doi: 10.1145/1029632.1029640 1, 2
- [6] I. S. MacKenzie and R. W. Soukoreff. Phrase sets for evaluating text entry techniques. In *CHI ’03 Extended Abstracts on Human Factors in Computing Systems*, CHI EA ’03, pp. 754–755. ACM, New York, NY, USA, 2003. doi: 10.1145/765891.765971 2
- [7] Z. Xu, Y. Meng, X. Bi, and X.-D. Yang. Phrase-gesture typing on smartphones. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–11, 2022. 1
- [8] C. Yu, Y. Gu, Z. Yang, X. Yi, H. Luo, and Y. Shi. Tap, dwell or gesture? exploring head-based text entry techniques for hmds. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, pp. 4479–4488, 2017. 1
- [9] D. Yu, K. Fan, H. Zhang, D. Monteiro, W. Xu, and H.-N. Liang. Pizzatext: Text entry for virtual reality systems using dual thumbsticks. *IEEE transactions on visualization and computer graphics*, 24(11):2927–2935, 2018. 1

¹<https://www.youtube.com/watch?v=YMORsQYLtU>